Stratified Cox Models

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Introduction

A stratified Cox model is the standard fix when the proportional-hazards assumption fails across the levels of a categorical covariate but holds within them. Stratification estimates a separate baseline hazard $h_{0s}(t)$ for each stratum while constraining the covariate effects β to be common across strata. The cost is that you cannot estimate the effect of the stratifier itself — its baselines are absorbed — but the gain is honest hazard-ratio estimation for the rest of the covariates without forcing PH across the problematic dimension.

Prerequisites

A working understanding of Cox proportional-hazards regression, the diagnostic role of Schoenfeld residuals, and the concept of stratification as an alternative to direct adjustment.

Theory

The stratified Cox model is

$$h(t \mid x, \text{stratum} = s) = h_{0s}(t) \exp(x^\top \beta).$$

The partial likelihood is summed across strata: each stratum contributes the standard Cox partial-likelihood term over its own risk sets, and the resulting score equations determine $\hat{\beta}$. Because the stratum baselines are unrestricted, no assumption is made about how $h_{0s}(t)$ relates across s. The trade-off is that the effect of being in stratum s versus another stratum cannot be estimated within the model — that contrast is precisely the term absorbed into the separate baselines.

Assumptions

Proportional hazards within each stratum (so the common β is meaningful), independent observations, and non-informative censoring.

R Implementation

```r
library(survival)

data(lung)
# If ph.ecog violates PH, stratify on it
fit <- coxph(Surv(time, status) ~ age + sex + strata(ph.ecog), data = lung)
summary(fit)

cox.zph(fit)
```

Output & Results

`summary(fit)` lists hazard ratios for non-stratified covariates with confidence intervals and Wald tests; the stratifier appears nowhere because its effect is absorbed into the per-stratum baselines. `cox.zph()` checks PH for the remaining covariates within strata.

Interpretation

A reporting sentence: "After Schoenfeld residuals indicated a PH violation for ECOG performance status (per-covariate $p < 0.01$), we fit a Cox model stratified on ECOG; the resulting hazard ratios for age (HR 1.02 per year, 95 % CI 1.01 to 1.03) and sex (HR 0.60 for female vs male, 95 % CI 0.42 to 0.85) are pooled across strata." When the stratifier is itself of clinical interest, supplement the model with stratum-specific Kaplan-Meier curves so readers can see the absorbed effect.

Practical Tips

- Stratify on a covariate when its PH fails and you do not need its hazard-ratio estimate; if you do need it, switch to a time-varying coefficient via `tt()` or `survSplit()` instead.
- Many small strata reduce the effective sample size for $\hat{\beta}$; pool levels (e.g. ECOG 0 vs 1+) when raw stratification leaves few events per stratum.
- Interactions between stratum and other covariates allow different β per stratum, generalising the model toward fully separate fits per stratum at the cost of degrees of freedom.
- Predictions and survival curves are stratum-specific; report which baseline is used for any predicted survival probability.
- Always verify that PH still holds for the unstratified covariates after stratification; failure there motivates additional stratification or a switch to time-varying effects.
- For continuous covariates that violate PH, stratification requires categorisation (typically into quartiles), which loses information; time-varying coefficients are usually preferable.

R Packages Used

`survival` for `coxph()` with `strata()`; `survminer::ggcoxzph()` for PH diagnostics within strata; `flexsurv` for parametric alternatives that can accommodate non-proportional hazards directly; `timereg` for additive hazards models when stratification is too coarse.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation